AHA COMMITTEE REPORT

A Reporting System on Patients Evaluated for Coronary Artery Disease

Report of the Ad Hoc Committee for Grading of Coronary Artery Disease, Council on Cardiovascular Surgery, American Heart Association

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INTRODUCTION

I give my enthusiastic support to the "Reporting System on Patients Evaluated for Coronary Artery Disease" which has been prepared by the committee of the Council on Cardiovascular Surgery chaired by Dr. Dwight McGoon. There is a great need for a standardized system of reporting and classifying patients, and much of the current uncertainty about the role of bypass surgery can be attributed to the lack of such a system.

The development of coding systems is always difficult. As the authors state in the preface, a balance must be struck between coding everything and having a form which is too long and on the other hand being too brief and too superficial. Another general principle relevant to coding systems is that you can't satisfy everyone, and there comes a time when you "must wrap it up and get it out." I have watched the progress of this committee for three years and know that a great deal of thinking has gone into this document. It is generally accepted as the best possible approach at this point in time, and the time has come to make it available for use. This system, or parts of it, has already been widely disseminated during the three years of its development, and components of this system have been incorporated into systems developed by other groups or institutions. The members of the committee can take satisfaction in the fact that their work has been copied by so many before it has been formally presented.

The American Heart Association has supported this project since its inception and, therefore, takes pride in sponsoring its first formal publication in the A.H.A. News Section of *Circulation*.

RICHARD S. ROSS, M.D.
A.H.A. Past President and Chairman,
Central Committee for
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PREFACE

There is a need for systematic accumulation of information on the many patients suspected of having disease of the coronary arteries. Any system for recording such information must strike a balance between being so concise as to be incomplete and being so complete as to lack conciseness. Because the selection of data to be collected is based on subjective evaluation and still incomplete information, the committee responsible for this reporting system — though it sought to approximate the ideal — recognizes that there is room for argument in some areas.

The committee has worked at this task over the past 3 years in the hope that a carefully designed system for recording relevant data on the many patients evaluated for coronary arterial disease would gain widespread use, either voluntarily or through the establishment of a national or international registry. This reporting system can be used to record subsequent as well as initial entries on each patient at regular anniversaries or when there is a change in his clinical status. Such a system will not only permit evaluation of the influence of surgical and medical methods of management on the natural history of this disease but also facilitate identification of both clinical and anatomic factors of prognostic importance in patients with coronary artery disease.

There are four parts to this system for recording relevant information on patients evaluated for disease of their coronary arteries: (1) the *Clinical and Arteriographic Report,* on which data relating to the patient are initially entered at the time of the first coronary arteriogram and which can be used each time a subsequent report of the status of the patient is made; (2) the *Surgical Report,* which is completed at the time of any cardiac operation; (3) the *Autopsy Report;* and (4) the *Booklet of Explanation and Instructions* for these three report forms.

The committee is also developing a method whereby these data can be recorded on mark-sense computer cards for appropriate storage and later collection (see Appendix).

The committee was assigned this task by the Executive Committee of the Council on Cardiovascular Surgery of the American Heart Association. Any merit that this system for reporting may have earned is the result of the multidisciplinary representation on this committee.

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INSTITUTION:
PATIENT NO.:
ENTRY NO.:

CLINICAL AND ARTERIOGRAPHIC REPORT

Patient Information

Name:
Address:
Telephone No.:
Social Sec. No.:
Age: Sex: M F
Weight: Height: BSA:
Referring Physician(s)
(Name, Address)
1.
2.
3.

Date of Entry:
Hospital No.:
Patient's Closest Contact
Name:
Address:
Telephone No.:
Relationship:

ASSOCIATED CONDITIONS

Cerebrovascular	YES	NO
Aortic	YES	NO
Renal	YES	NO
Peripheral Vascular	YES	NO
Chronic Pulmonary	YES	NO
Valvular Heart Disease	YES	NO
AS ☐ AR ☐ MS ☐ MR ☐		
Family History	YES	NO
Diabetes	YES	NO
Systemic Hypertension	YES	NO
Hyperuricemia	YES	NO
Hypercholesterolemia	YES	NO
Hypertriglyceridemia	YES	NO
IIa ☐ IIb ☐ III ☐ IV ☐		
Smoking (Cigarettes)	YES	NO
Pack-Yrs ———		
½/Day ☐ Quit ☐		
Other Heart Disease	YES	NO

PRESENT THERAPY

Nitroglycerin	YES	NO
Propranolol	YES	NO
Antilipemic Drugs	YES	NO
Low-fat Diet	YES	NO
Digitalis	YES	NO
Diuretics	YES	NO
Nitrates, Medium- or Long-acting	YES	NO
Antiarrhythmic Drugs	YES	NO
Antihypertensive Drugs	YES	NO
Supervised, Graded Exercise	YES	NO
Other Form of Therapy (—————)	YES	NO

ANGINA AND ANGINOID PAIN

Chest Pain	YES	NO
Angina	YES	NO
Unstable Angina	YES	NO
New Angina of Effort	☐	
Changing Pattern	☐	
New Angina at Rest	☐	
Chest Pain, Uncertain Etiology	YES	NO
Severity of Chest Pain		
Mild	☐	
Moderate	☐	
Severe	☐	

Chest Pain Relationships
Effort ☐ Rest (Awake) ☐
Emotion ☐ Nocturnal ☐
Present Trend of Chest Pain
Decreasing ☐ Increasing ☐
Stable ☐ On Therapy ☐
History of Chest Pain, None at Present ☐
Duration of Present Chest Pain ——— Mo.
Time Since First Episode of Pain ——— Mo.

CONGESTIVE HEART FAILURE (CHF)

CHF at This Time

None ☐ Mild ☐ Moderate ☐

Severe ☐ Cardiogenic Shock ☐

Present Trend of CHF

Decreasing ☐ Increasing ☐

Stable ☐ On Therapy ☐

Duration of Present CHF ——— Mo.

Past History of CHF (Patient Now Compensated) YES NO

Time Since First Episode of CHF ——— Mo.

Chest X-ray

Normal Heart Size ☐ Cardiomegaly ☐

MYOCARDIAL INFARCTION (MI)

History or Evidence of Myocardial Infarction YES NO

Number Documented Infarctions 1 ☐ 2 ☐ 3 ☐ >3 ☐

Complications of Any Infarction

Congestive Failure ☐ Cardiogenic Shock ☐

Serious Arrhythmia ☐

Time Since Last MI ——— Mo.

Time Since First MI ——— Mo.

PATIENT'S ACTIVITIES

Employment

Laborer ☐ Clerical ☐ Professional ☐ Homemaker ☐

Part-Time ☐ Retired ☐

Ordinary Stress ☐ Severe Stress ☐

Quit Work ☐ Reduction in Work ☐

Because of Cardiac Disease ☐ On Physician's Advice ☐

Hobbies

Light Hobbies ☐ Moderate Activities ☐ Athletic Hobbies ☐

Abandoned ☐ Curtailed ☐

Because of Cardiac Symptoms ☐ On Physician's Advice ☐

PREVIOUS CARDIAC OPERATION

Vein Bypass Graft	YES	NO
IMA Bypass Graft	YES	NO
Other Arterial Bypass	YES	NO
Internal Mammary Implant	YES	NO
Aortic Valve Operation	YES	NO
Mitral Valve Repair	YES	NO

Date of Operation ————

Mitral Valve Replacement	YES	NO
Resection Ventricular Segment	YES	NO
Poudrage or Abrasion	YES	NO
Other Cardiac Operation	YES	NO

Type ————

ELECTROCARDIOGRAM Normal: YES NO

(Check Appropriate Box)	Present YES	Present NO	Site of Changes: Anterolateral (I, aVL, V_6)	Inferior (II, III, aVF)	Anteroseptal (V_1-V_5)
Infarction (Q & QS Pattern)					
ST Junction and Segment Depression					
T-Wave Abnormality					
ST Segment Elevation					

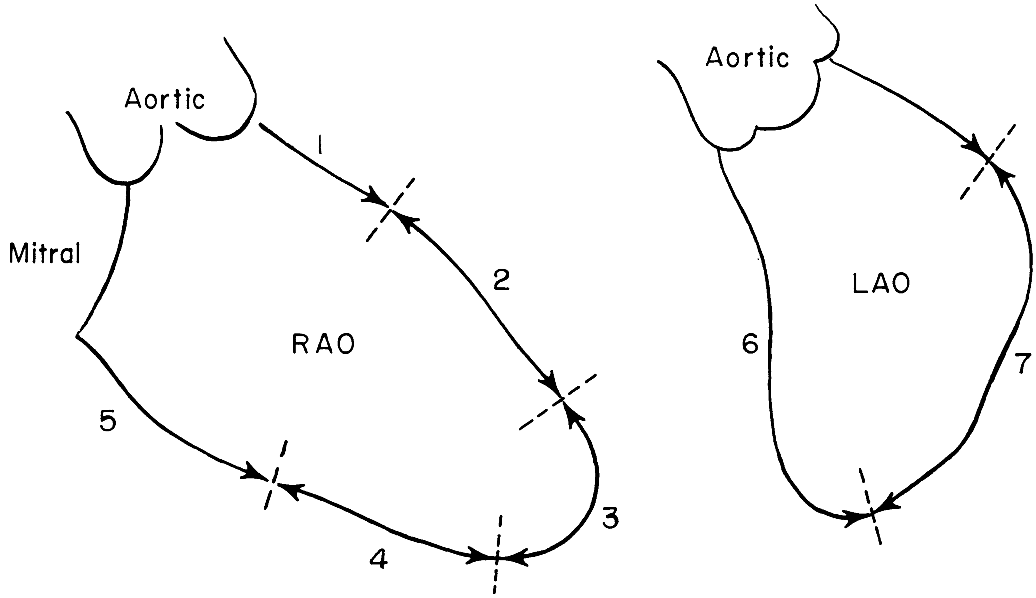
QRS Axis Deviation YES NO
Left ☐ Right ☐ Extreme ☐ Indeterminate ☐
Ventricular Hypertrophy (VH) (High-Amplitude R Waves) YES NO
LVH ☐ RVH ☐
Atrioventricular (A-V) Conduction Defect YES NO
Complete (3°) ☐ Partial (2°) ☐ P-R$\geq$0.22 sec ☐
WPW ☐ Short P-R ☐ Pacemaker ☐
Ventricular Conduction Defect YES NO
RBB ☐ LBB ☐ LAH ☐ LPH ☐
Arrhythmia YES NO
Occasional PVC ☐ Frequent PVC ☐ Multifocal PVC ☐
Ventricular Tachycardia ☐ Idioventricular ☐
Atrial Fibrillation or Flutter ☐ Supraventricular Tachycardia ☐
Sinus Bradycardia ☐ Nodal ☐ Other 4ther ________
Miscellaneous: Low-Amplitude YES NO
QRS Transition Zone: V_2 ☐ V_3 ☐ V_4 ☐ V_5 ☐ V_6 ☐

ELECTROCARDIOGRAPHIC STRESS TEST Done: YES NO

	Heart Rate Achieved	
	Positive ECG	Negative ECG
Exercise		
Atrial Pacing		

Typical Angina Pectoris Induced YES NO
Other ________ YES NO

LEFT VENTRICULOGRAM (LVG)



Propranolol YES NO
Nitroglycerin YES NO

Relationship of LVG to Coronary Arteriogram
Preceded ☐
Followed ☐

Contractility

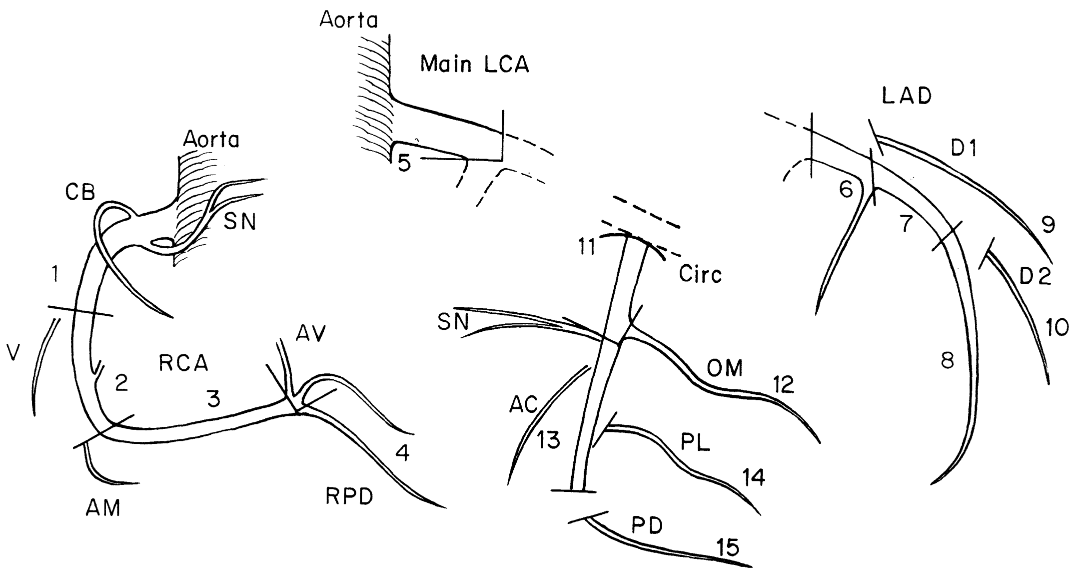
Segment	Normal	Reduced	None	Dyskinetic	Aneurysmal	Undefined
1. Anterobasal						
2. Anterolateral						
3. Apical						
4. Diaphragmatic						
5. Posterobasal						
6. Septal Wall						
7. Posterolateral						

HEMODYNAMICS

Aortic Pressure (Syst/Diast) ———/——— mm Hg Heart Rate ———/min
LV Pressure (Syst/LVEDP) ———/——— mm Hg Cardiac Index ——— liters/min/m^2
LV End-Diastolic Volume ——— ml/m^2 BSA Ejection Fraction ———%
VSD YES NO Pul/Systemic Flow Ratio ———

Mitral Valve Gradient ——— mm Hg Aortic Valve Gradient ——— mm Hg
Mitral Regurgitation: None ☐ Slight ☐ Moderate ☐ Severe ☐
Aortic Regurgitation: None ☐ Slight ☐ Moderate ☐ Severe ☐ No Aortogram ☐

CORONARY ARTERIOGRAM



Branch	Normal	Gives Collaterals	Small	25%	50%	75%	90%	Filled By Collaterals	99%	100%	Graftable	Absent
RCA 1												
2												
3												
4												
LCA 5												
LAD 6												
7												
8												
9												
10												
CIRC 11												
12												
13												
14												
15												

ANGIOGRAPHY OF CORONARY BYPASS GRAFT

Insertion Segment of Bypass Graft		Patent	Occluded	Normal	Irregularly Narrowed (< 50%)	Diffusely Narrowed (> 50%)	Aneurysmal	Diameter Most Severe Narrowing					Selective Injection	Unknown
								< 1 mm (90%)	1 to < 2 mm (75%)	2 to < 3 mm (50%)	3 to < 4 mm (25%)	> 4 mm (normal)		
RCA	1													
	2													
	3													
	4													
LCA	5													
LAD	6													
	7													
	8													
	9													
	10													
CIRC	11													
	12													
	13													
	14													
	15													

RECOMMENDED MEDICAL THERAPY

- Nitroglycerin ☐
- Propranolol ☐
- Antilipemic Drugs ☐
- Low-fat Diet ☐
- Digitalis ☐
- Diuretics ☐
- Nitrates, Medium- or Long-acting ☐
- Antiarrhythmic Drugs ☐
- Antihypertensive Drugs ☐
- Supervised, Graded Exercise ☐
- Other Form of Therapy (————————) ☐
- No Medical Therapy Recommended ☐

RECOMMENDED SURGICAL THERAPY

- Operation Recommended ☐
- Operation Recommended, Patient Refused ☐
- Coronary Artery Operation Performed ☐
- Operation Not Recommended ☐

PATIENT DEATH YES NO

Autopsy Performed YES NO

INSTITUTION:
PATIENT NO.:
ENTRY NO.:

SURGICAL REPORT
Patient Information

Name:
Social Sec. No.:
Date of Operation:

Previous Entry Applies ☐
Accompanied by New Entry ☐

Surgical Information
Primary Surgeon:
Staff ☐ Resident ☐

Date of Coronary Arteriogram ————

OPERATIVE PROCEDURE

Insertion Segment of Bypass Graft		Type/Origin	Endarterectomy	Quality Coronary Artery (Normal, Diseased, Occluded)	Diameter Coronary Lumen (mm)	Aortic Cross-Clamp Time (min)	Graft Flow (ml/min)	Unusual Problem
RCA	1							
	2							
	3							
	4							
LCA	5							
LAD	6							
	7							
	8							
	9							
	10							
CIRC	11							
	12							
	13							
	14							
	15							

Type of Graft
1 - Saph. vein autograft
2 - Other venous autograft
3 - Right IMA
4 - Left IMA
5 - Other arterial autograft
6 - Venous homograft
7 - Arterial homograft
8 - Prosthetic artery
9 - Arterial heterograft
10 - Other————

Origin of Graft
A - Aorta
B - Natural origin
C - Side branch
D - Continuation
E - Other————

Ventricular Fibrillation
None ☐ Induced ☐ Spontaneous ☐ Countershock Required ☐
Temperature Myocardial Perfusate
Normothermia ☐ Moderate Hypothermia (28–35°C) ☐
Deep Hypothermia (< 28° C) ☐
Surface Cardiac Cooling YES NO
Duration of Extracorporeal Circulation ________Min.

OTHER CARDIAC PROCEDURES
Valve Surgery
Aortic Valve Replaced ☐ Repaired ☐
Mitral Valve Replaced ☐ Repaired ☐
Ventricular Segment
Resected ☐ Plicated ☐ Segment No. ________
Resected Segment
Aneurysm ☐ Acute Infarct ☐ Healing Infarct ☐ Healed Infarct ☐ Fibrosis ☐
Ventricular Septal Defect Repair YES NO
Internal Mammary Artery Implant YES NO Segment No. ________
Other Operative Procedures ____

POSTOPERATIVE COURSE No Complication ☐ Complication ☐
Electrocardiographic Change YES NO

Change From Preoperative ECG (Check Appropriate Box)			Site of Changes		
	YES	NO	Anterolateral (I, aVL, V_6)	Inferior (II, III, aVF)	Anteroseptal (V_1-V_6)
New Q or QS Pattern					
New ST Segment Shift					
New T-Wave Inversion					

Serum Enzyme Determinations

Serum Enzyme	Upper Limit of Institution Normal*	Maximum Postoperative Value*	Not Determined
SGOT			
LDH			
Total CPK			
CPK-Isozyme			

*U/liter.

Morbidity	YES	NO
Mediastinal Infection		
Pulmonary Embolism		
Low Cardiac Output		
Excessive Hemorrhage		
Reoperation for Hemorrhage		
Arrhythmia		
Other————		

Arrhythmia
- ☐ Atrial Fibrillation or Flutter
- ☐ Supraventricular Tachycardia
- ☐ Frequent PVC
- ☐ VT/VF
- ☐ Other————

Hospital Mortality YES NO
- ☐ Electrical Instability
- ☐ Myocardial Infarction
- ☐ Infection

Days After Operation————
- ☐ Low Cardiac Output
- ☐ Hemorrhage
- ☐ Other————

Autopsy Performed YES NO

INSTITUTION:
PATIENT NO.:
ENTRY NO.:

Name:
Social Sec. No.:
Date of Autopsy:

AUTOPSY REPORT
Patient Information

LEFT VENTRICLE

Pathologic Information

Segment	Normal	Infarct Acute (< 1 wk)	Infarct Healing (1 wk to 1 mo)	Infarct Healed (> 1 mo)	Trans-mural	Subendo-cardial	Fibrosis	Aneurysm
1. Antero-basal								
2. Antero-lateral								
3. Apical								
4. Diaphrag-matic								
5. Postero-basal								
6. Septal Wall								
7. Postero-lateral								

OTHER FINDINGS

Ventricular Septal Defect YES NO

Aortic Valve Stenosis □ Regurgitation □
Mitral Valve Stenosis □ Regurgitation □
Heart Weight ———— g
Lung Weight ———— g
Fibrous Pericarditis YES NO
Mediastinum Infection □
Hematoma □
Normal □

CORONARY ARTERIES

Postmortem Coronary Arteriogram YES NO

Branch	Normal	Gives Collaterals*	Small	25%	50%	75%	90%	Filled by Collaterals*	99%	100%	Graftable*	Absent
RCA 1												
2												
3												
4												
LCA 5												
LAD 6												
7												
8												
9												
10												
CIRC 11												
12												
13												
14												
15												

*Complete only if postmortem coronary arteriogram done.

CONDITION OF CORONARY ARTERIES

Insertion Segment of Bypass Graft		Patent	Occluded	Normal	Irregularly Narrowed (< 50%)	Diffusely Narrowed (> 50%)	Aneurysmal	Diameter Most Severe Narrowing				
								< 1 mm (90%)	1 to < 2 mm (75%)	2 to < 3 mm (50%)	3 to < 4 mm (25%)	> 4 mm (normal)
RCA	1											
	2											
	3											
	4											
LCA	5											
LAD	6											
	7											
	8											
	9											
	10											
CIRC	11											
	12											
	13											
	14											
	15											

CAUSE OF DEATH

Classification
- Cardiac, Coronary ☐
- Cardiac, Non-coronary ☐
- Non-cardiac ☐

State Underlying Cause________________________

CLINICAL AND ARTERIOGRAPHIC REPORT

Explanation and Instructions

General Instructions

Data will be recorded by: (1) writing in the requested information (name, address, referring physician, etc.), (2) inserting an "X" in the appropriate box if the respective situation or condition applies to the patient, or (3) circling "YES" or "NO" to answer the question asked. If no data are recorded, or if the question is not answered, it will be understood that this information is unknown or indeterminate.

Patient Identification Number

A patient identification number is given to that patient by the participating institution. *Note:* Each patient on entering the study is assigned an identification number that will always remain the same, for that hospital.

Entry Number

The entry number refers to the specific occasion on which information on a patient is being recorded. The first report should be marked as *First Entry* and should bear the date of the first coronary arteriogram. All data recorded on the first coronary arteriogram should reflect the status of the patient at the time of the first coronary arteriogram.

Subsequent entries, whether or not they include information about a coronary arteriogram, should be recorded only on the pertinent portions of the report. The appropriate sequential entry number should be recorded and the date on which the information for that entry was obtained should be marked. It is anticipated that annual follow-up entries will be made on each patient. Reasons for additional entries other than the annual entry include a change in symptomatic or functional status, the occurrence of an acute myocardial infarction, follow-up coronary arteriogram, cardiac operation, and death.

Patient Information

Enter the patient's name, address, telephone number, and social security number; the name and telephone number of the patient's closest contact; and the name(s) and office address(es) of referring physicians.

Record the date of the entry (e.g., if reason for entry was cardiac catheterization and coronary arteriogram, or annual examination, the date of entry is the date that this was done), the patient's hospital number, his/her age, sex, body weight (in kilograms), height (in centimeters), and body surface area (in m²).

Referring Physician

The physician(s) who initiated the patient's investigation and who probably will remain aware of the patient's continued course is (are) termed the referring physician(s).

Specific Instructions

Associated Conditions

Cerebrovascular

Cerebrovascular disease: Obstructive or occlusive involvement of the carotid, vertebral, innominate arteries, or their intracranial branches, that is associated with, or likely to cause, cerebral ischemia.

YES = Above present on basis of specific documentation such as arteriography or clear-cut clinical findings, such as history of a stroke or transient cerebral ischemia, or evidence of a residual neurologic deficit of a previous stroke.

NO = None of above clearly present.

Aortic

Aortic disease: Atherosclerotic involvement, cystic medionecrosis, or other disease resulting in aortic aneurysm, dissection, marked dilatation, obstruction, occlusion, or inherent weakness of the aortic vessel wall.

YES = Above present on basis of specific documentation or clear-cut clinical findings.

NO = None of above clearly present.

Renal

Renovascular disease or parenchymal involvement of the kidneys resulting in measurable impairment of renal function.

YES = Above established by renal arteriography and/or unquestionable history and/or clearly abnormal clinical chemistry values.

NO = None of above clearly present.

Peripheral Vascular

Obstructions or occlusions of peripheral arteries.

YES = Established by arteriography or clear-cut clinical findings.

NO = None of above clearly present.

Chronic Pulmonary

Chronic pulmonary disease: Emphysema, chronic bronchitis, bronchiectasis, pulmonary fibrosis, symptomatic pneumoconiosis, pulmonary hypertension, granulomatous or neoplastic disease, or kyphoscoliosis.

YES = Diagnosis by thoracic roentgenography,

pulmonary function studies, or specific testing.
NO = None of above clearly present.

Valvular Heart Disease

Aortic stenosis, aortic regurgitation, mitral stenosis, or mitral regurgitation, regardless of etiology.

NO = No clinical evidence of valvular heart disease.

"AS" = Aortic stenosis documented by invasive diagnostic studies or clear-cut clinical findings.

"AR" = Aortic regurgitation documented by invasive diagnostic studies or clear-cut clinical findings.

"MS" = Mitral stenosis documented by invasive diagnostic studies or clear-cut clinical findings.

"MR" = Mitral regurgitation documented by invasive diagnostic studies or clear-cut clinical findings. (Quantification of above valvular involvement to be recorded under Hemodynamics section.)

Family History

Angina pectoris or myocardial infarction before age 55 in any of patient's parents, siblings, aunts, or uncles.

Diabetes

Abnormal blood glucose: Either 2-hour postprandial hyperglycemia or a glucose tolerance result above normal range for participating institution.

Systemic Hypertension

Diastolic blood pressure 95 mm Hg or above at time of entry or medical history of such an elevation.

Hyperuricemia

Serum uric acid above normal range for participating institution.

Hypercholesterolemia

Serum levels of cholesterol above normal range for participating institution.

Hypertriglyceridemia

Serum levels of triglycerides above normal range for participating institution. IIa, IIb, III, IV: If available and appropriate, score also hyperlipidemia type IIa, IIb, III, IV.

Smoking

Cigarette-smoking history.

YES = Patient is smoking cigarettes at present.

NO = Patient *never* smoked cigarettes. *Note:* If this is circled, disregard other entries concerning smoking history.

"Pack-yrs" = Cigarette pack-year: Score total number of pack-years smoked. This is calculated by multiplying packs smoked per day by the number of years the patient smoked.

"½/Day" = History of cigarette smoking but never exceeding one-half pack of cigarettes per day at any time.

"Quit" = Patient smoked in past and no longer is smoking cigarettes. (Also indicate "pack-yrs" above.)

Other Heart Disease

YES = If patient has or had heart disease other than those listed, please record appropriate diagnosis or diagnoses.

NO = If patient has or had no other heart disease.

Present Therapy

Indicate whether the patient is on any of the following medications at the time of this entry:
Nitroglycerin
Propranolol
Digitalis
Diuretics
Antilipemic Drugs
Low-fat Diet
Nitrates, Medium- or Long-acting
Antiarrhythmic Drugs
Antihypertensive Drugs
Supervised, Graded Exercise
Other Form of Therapy (please list)

Note: Under antihypertensive medications, include all medications except diuretic agents, which are to be entered separately.

Angina and Anginoid Pain

Chest Pain

At the time of this entry, any form of chest pain that is severe or frequent enough to require a specific therapeutic regimen or referral for coronary arteriography.

YES = Chest pain is present.

NO = No complaint of chest pain whatsoever.

Angina

Pain usually arising in chest predictably induced by exertion — whether or not it has occurred at other times — and relieved by rest and/or nitroglycerin within approximately 5 minutes. No ECG evidence of recent myocardial infarction and no serum enzyme changes.

YES = Angina is present.

NO = Angina is not present.

Unstable Angina

Described symptoms of chest pain began within 3 weeks of hospitalization, the last episode of chest pain having occurred within 1 week before this entry. No ECG change reflecting recent myocardial infarction, no elevation of serum myocardial enzyme levels. (It is recognized that more than one entry may be circled under this section.)

YES = Unstable angina pectoris is present.

NO = Unstable angina pectoris is not present.

"*New Angina of Effort*" = The recent onset of angina of effort; onset of angina for the first time or the recurrence of angina after being free of angina for at least 6 months.

"*Changing Pattern*" = Angina of effort with changing pattern. The patient with previously stable angina of effort whose pain has worsened or intensified so that there is an increase in the frequency, duration, severity, ease of provocation, or radiation; also, usually less responsive to sublingual nitroglycerin.

"*New Angina at Rest*" = Occurring at rest, may last more than 15 minutes, and may not be relieved by nitroglycerin. Often accompanied by transient ST segment change (elevation or depression) or T-wave inversion with pain.

Chest Pain, Uncertain Etiology

Any chest pain that is atypical of angina and believed not to result from ischemic heart disease.

YES = Chest pain of uncertain etiology is present.

NO = Chest pain of uncertain etiology is not present.

Severity of Chest Pain

Severity of chest pain at the time of this report.

"*Mild*" = Mild chest pain: "Ordinary" physical activity or emotional stress (i.e., that level of physical or emotional stress which can be considered usual and typical of everyday activity for that individual patient) does not cause chest pain. Chest pain can be elicited by effort that is greater than ordinary efforts. Examples: running for a bus, pushing a car.

"*Moderate*" = Moderately severe chest pain: Ordinary physical or emotional stress, typical of everyday activity for the individual, can cause chest pain. Examples: climbing stairs, playing golf with a cart, walking briskly uphill.

"*Severe*" = Severe chest pain: In addition to being severely limited in the physical activity he can perform, patient has angina at rest (either decubitus or nocturnal). Synonymous with NYHA class IV functional status.

Chest Pain Relationships

Chest pain that characteristically is induced by the following activities or states:

"*Effort*" = Effort-induced pain.

"*Emotion*" = Emotion-induced pain.

"*Rest*" = Pain occurring at rest, awake.

"*Nocturnal*" = Nocturnal pain.

Present Trend of Chest Pain

"*Decreasing*" = Severity, duration, or frequency of pain, or ease of provocation is decreasing.

"*Stable*" = No change in symptoms — i.e., a stable pattern.

"*Increasing*" = Severity, duration, or frequency of pain, or ease of provocation is increasing.

"*On Therapy*" = May be checked in conjunction with boxes for above to indicate, for instance, that symptoms are decreasing in severity with therapy or are stable or are increasing in severity despite therapy.

History of Chest Pain

History of chest pain, but none at the present time. Answer only if "No" is checked in response to the first question of this section: "Is chest pain present at the time of this entry?"

Duration of Present Chest Pain

Record the duration (in months) of severity of chest pain at time of this report.

Time Since First Episode of Pain

Record the interval (in months) from first episode of chest pain. (Could be same period as for "Duration" if severity has been at a rather steady level since onset.)

Congestive Heart Failure (CHF)

CHF at This Time

Congestive heart failure at the time of this report.

"*None*" = No evidence of heart failure at time of this entry. Does not exclude previous episodes of heart failure.

"*Mild*" = Patient has slight limitation of physical activity. Patient comfortable at rest. Ordinary physical activity results in fatigue, palpitations, or dyspnea.

"*Moderate*" = Patient has moderate to marked limitation of physical activity. Patient comfortable at rest. Less than ordinary physical activity causes fatigue, palpitations, or dyspnea.

"*Severe*" = Inability to carry on any physical activity without fatigue and dyspnea. In addition, patient has either orthopnea or paroxysmal nocturnal dyspnea. (Some or all of the following physical findings may be present: tachycardia, third and fourth heart sounds, cyanosis, neck vein distention, cardiomegaly, rales, pleural effusion, hepatosplenomegaly, and pulmonary edema.)

"*Cardiogenic Shock*" = Sudden decrease of cardiac output following acute myocardial infarction resulting in inadequate perfusion to major organ-systems. Cardiogenic shock is said to be present when associated with each of the following:*

(1) Systolic pressure <90 mm Hg or, in the previously hypertensive patient, 80 mm Hg below documented prior basal systolic level; and

*Myocardial Infarction Research Unit Classification, May 1973.

(2) Cardiac index <2.2 liters/min/m² and PCW >15 mm Hg or LVEDP >15 mm Hg; and
(3) Evidence of decreased arterial perfusion of two or three vascular beds: renal (urine output <20 ml/hr), cerebral (mental confusion or coma), and cutaneous (cyanosis or vasoconstriction of extremities, coolness, and diaphoresis); and
(4) Treatment of hypoxemia, acidemia, and serious dysrhythmia has been effected.

Present Trend of CHF

Present trend of heart failure.

"*Decreasing*" = Severity, duration, or frequency of failure, or ease of provocation is decreasing.

"*Stable*" = No change in symptoms — i.e., stable pattern.

"*Increasing*" = Severity, duration, or frequency of failure, or ease of provocation is increasing.

"*On Therapy*" = On therapy: May be checked in conjunction with the above boxes to indicate, for instance, that symptoms are decreasing with therapy or are stable or are increasing despite therapy.

Duration of CHF

Record duration of present level of severity of congestive heart failure in months.

Past History of CHF

Past history of congestive heart failure, patient now compensated. (Answer only if "None" checked above.)

YES = Patient has past history of congestive heart failure but is not presently in failure.

NO = Patient has never been in congestive heart failure.

"*Time Since First Episode of CHF*" = Time since first episode of congestive heart failure, recorded in months (could be the same as for duration of present heart failure if severity of failure has changed little since onset).

Chest X-ray

"*Normal Heart Size*"

"*Cardiomegaly*" = Cardiomegaly is evident; cardiothoracic ratio >50%.

Myocardial Infarction (MI)

History or Evidence of MI

History of or evidence of myocardial infarction, when myocardial infarction is defined as ischemic necrosis of myocardial tissue. Evidence of myocardial infarction is defined as localized abnormality in depolarization of ECG with abnormal Q waves (or loss of R waves) and/or diagnostic changes in concentrations of serum myocardial enzymes.

YES = Patient has history or ECG evidence of acute myocardial infarction.

NO = No history or evidence of acute myocardial infarction.

Number of Documented Infarctions

"*Number*" = Number of documented infarctions. Record number of documented myocardial infarctions in the past: one, two, three, or greater than three.

Complications of Any Infarction

Indicate if any myocardial infarction has been complicated by one of the following:

"*Congestive Failure*" = See definition under Congestive Heart Failure section.

"*Cardiogenic Shock*" = See definition under Congestive Heart Failure section.

"*Serious Arrhythmia*" = Ventricular tachycardia, ventricular fibrillation, complete A-V block, ventricular asystole.

Time Since Last MI

Time since last myocardial infarction. Record time in months since last documented myocardial infarction.

Time Since First MI

Time since first documented myocardial infarction. Record time in months since first documented myocardial infarction.

Patient's Activities

Note: All of the following activities, professions, and hobbies refer to those of the patient's normal living pattern prior to any changes required by ischemic heart disease.

Employment

"*Laborer*" = Routine performance in regular working day of heavy manual labor and physical exertion. Examples: lifting, climbing, pushing heavy items.

"*Clerical*" = Any form of employment not regularly associated with heavy manual labor and not involving stressful, decision-making responsibility. Examples: work of tellers, cashiers, teachers, precision mechanics, electronic technicians, secretaries, typists, salesmen.

"*Professional*" = Occupation requiring advanced training and involving mental rather than physical work, as well as definite amount of stress and responsibility, associated with decision making. Examples: all executive positions in industrial, political, military, educational, and commercial enterprises plus positions in professions involving the security of others, such as physicians, lawyers, pilots, airport tower operators.

"*Homemaker*" = Work in the home, including chores and responsibilities of managing the household including cleaning and making beds.

"*Part-Time*" = Predominant work habit of less than 40 hours per week (if not scored, a 40-hour work-week or longer is implied).

"*Retired*" = Discontinued work on reaching retirement age, as opposed to quitting because of doctor's advice, incapacitation, or fear.

"*Ordinary Stress*" = Ordinary stress, either physical or mental, as associated with any of the above work categories.

"*Severe Stress*" = Severe stress, either physical or mental, as associated with any of the above work categories.

"*Quit Work*" = Quit before retirement for some reason.

"*Reduction in Work*" = Indicates reduction in time spent at work, as working fewer hours or avoiding strenuous aspect of work. (This is a recent change as opposed to "Part-Time" category above.)

"*Because of Cardiac Disease*" = Patient has been forced to quit or reduce working because of cardiac symptoms prior to retirement age (if also recommended by a physician, "On Physician's Advice" additionally is scored).

"*On Physician's Advice*" = Reduction or cessation of work on a physician's advice.

Hobbies

"*Light Hobbies*" = Refers to activities performed for pleasure and relaxation, involving only slight physical activity. Examples: photography, light carpentry, strolling, fishing, light gardening.

"*Moderate Activities*" = Indicative of activities performed for pleasure and relaxation without ideas of competition, endurance, or excellence in them. Examples: golfing, gardening, jogging, bowling.

"*Athletic Hobbies*" = Physically demanding recreational activities usually involving competition, endurance, and/or team effort. Examples: football, basketball, soccer, tennis, squash, hockey, skiing.

"*Abandoned*" = Hobby activity has been stopped for some reason.

"*Curtailed*" = Hobby activity has been restricted, indicating that patient reduced the time or effort, or gave up a more strenuous hobby in favor of a less demanding one.

"*Because of Cardiac Symptoms*" = Patient has been forced to abandon or curtail hobby because of symptoms (if physician advised same, "On Physician's Advice" should also be scored).

"*On Physician's Advice*" = Patient abandoned or curtailed hobby activity on physician's advice.

PREVIOUS CARDIAC OPERATION

This section is completed only on the first entry for this patient. If cardiac operation is performed subsequent to the first entry, the Surgical Entry Form will be used to record the operative details in this patient's file. This section is justified on the basis that the first Clinical and Arteriographic Report entered on this patient will be dated at the time of the first coronary arteriogram for which the documentation of the results is available and that details of any operation preceding the date of the first entry will therefore also be less than adequately documented.

"Vein Bypass Graft"	= Saphenous vein bypass graft.
"IMA Bypass Graft"	= Internal mammary artery bypass graft.
"Other Arterial Bypass"	= Other type of coronary arterial bypass graft.
"Internal Mammary Implant"	= Internal mammary artery implant (Vineberg procedure or modification).
"Aortic Valve Operation"	= Aortic commissurotomy or aortic valve replacement.
"Mitral Valve Repair"	= Includes mitral commissurotomy or annuloplasty or any repair of mitral valve not involving prosthetic valve replacement.
"Mitral Valve Replacement"	= Prosthetic mitral valve replacement.
"Resection Ventricular Segment"	= Resection of aneurysm or scar or infarct or any segment of ventricular myocardium.
"Poudrage or Abrasion"	= Pericardial poudrage or epicardial abrasion.
"Other Cardiac Operation"	= Other cardiac operation not listed above. Please write in procedure performed.

ELECTROCARDIOGRAM*

Normal

YES = If resting electrocardiogram is within normal limits, disregard all other resting electrocardiographic scoring possibilities.

NO = If electrocardiogram abnormal, circle NO *and* check additional scoring possibilities as applicable.

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*Adapted from: G. A. Rose and H. Blackburn, "Electrocardiographic Reading Codes" (Appendix D) in "The Coronary Drug Project: Designed, Methods and Baseline Results," Circulation 47 Suppl **1**:39-42, 1973. (Minnesota Code). The committee wishes to acknowledge its indebtedness to Drs. Rose and Blackburn for the great assistance their foregoing work has provided.

Infarction (Q & QS Pattern)

(Do not code in presence of Wolff-Parkinson-White [WPW] syndrome or complete left bundle-branch block.)

YES = Electrocardiographic evidence of myocardial infarction based on any one of the Q and QS patterns described in this section. Indicate whether location of infarction is anterolateral, inferior, or anteroseptal.

NO = No Q or QS pattern present reflecting myocardial infarction.

(1) Anterolateral site (leads I, aVL, V_6):
- Q/R amplitude ratio 1/3 or more plus Q duration 0.03 sec or more in any of leads I, V_6.
- Q duration 0.04 sec or more in any of leads I, V_6.
- Q duration 0.04 sec or more plus R amplitude of 3 mm* or more in lead aVL.
- Q/R amplitude ratio 1/3 or more plus Q duration at least 0.02 sec and less than 0.03 sec in any of leads I, V_6.
- Q duration at least 0.03 sec and less than 0.04 sec in any of leads I, V_6.
- Q/R amplitude ratio at least 1/5 and less than 1/3 plus Q duration at least 0.02 sec and less than 0.03 sec in any of leads I, V_6.
- Q duration at least 0.03 sec and less than 0.04 sec plus R amplitude of 3 mm or more in lead aVL.

(2) Posterior (inferior) site (leads II, III, aVF):
- Q/R amplitude ratio 1/3 or more plus Q duration 0.03 sec or more in lead II.
- Q duration 0.04 sec or more in lead II.
- Q duration 0.05 sec or more in lead III plus any Q wave at least 1.0-mm amplitude in aVF.
- Q duration 0.05 sec or more in lead aVF.
- Q/R amplitude ratio 1/3 or more plus Q duration at least 0.02 sec and less than 0.03 in lead II.
- Q duration at least 0.03 sec and less than 0.04 sec in lead II.
- QS pattern in lead II.
- Q duration at least 0.04 sec and less than 0.05 sec in lead III plus any Q wave at least 1.0-mm amplitude in aVF.
- Q duration at least 0.04 sec and less than 0.05 sec in lead aVF.
- Q amplitude of 5 mm or more in either of leads III and aVF.
- Q/R amplitude ratio at least 1/5 and less than 1/3 plus Q duration of at least 0.02 sec and less than 0.03 sec in lead II.
- Q duration at least 0.03 sec and less than 0.04 sec in lead III plus any Q wave at least 1.0-mm amplitude in lead aVF.
- Q duration at least 0.03 sec and less than 0.04 sec in lead aVF.
- QS pattern in each of leads III and aVF.

(3) Anteroseptal site (leads V_1–V_5):
- Q/R amplitude ratio 1/3 or more plus Q duration 0.03 sec or more in any of leads V_2–V_5.
- Q duration 0.04 sec or more in any of leads V_1–V_5.
- QS pattern when R wave is present in adjacent lead to the right on the chest in any of leads V_2–V_6.
- QS pattern in all leads V_1–V_4, V_1–V_5, or V_1–V_6.
- Q/R amplitude ratio 1/3 or more plus Q duration at least 0.02 sec and less than 0.03 sec in any of leads V_2–V_5.
- Q duration at least 0.03 sec and less than 0.04 sec in any of leads V_2–V_5.
- QS pattern in all of leads V_1–V_3.
- R amplitude decreasing to 2 mm or less and absence of RVH, complete RBBB, and incomplete RBBB between any of leads V_2 and V_3, V_3 and V_4, and V_4 and V_5.
- Q/R amplitude ratio at least 1/5 and less than 1/3 plus Q duration at least 0.02 sec and less than 0.03 sec in any of leads V_2–V_5.
- QS pattern in absence of LVH in each of leads V_1 and V_2.

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*It is assumed throughout that 1 mm = 0.1 mv.

ST Junction and Segment Depression

(Do not code in the presence of WPW syndrome, complete left bundle-branch block, complete right bundle-branch block, or interventricular block, with QRS duration of 0.12 sec or more in any of leads I, II, III, aVL, aVF.)

YES = If any of the ST junction or ST segment depression abnormalities described within section are present. Indicate site of ECG change (anterolateral, inferior, or anteroseptal).

NO = If no ST junction or ST segment depression is present.

(1) Anterolateral site (leads I, aVL, V_6):

ST-J depression 1.0 mm or more and ST segment horizontal or downward-sloping in any of leads I, aVL, V_6.

ST-J depression at least 0.5 mm and less than 1.0 mm and ST segment horizontal or downward-sloping in any of leads I, aVL, V_6.

No ST-J depression as much as 0.5 mm but ST segment downward-sloping and segment or T-wave nadir at least 0.5 mm below P-R baseline in any of leads I, aVL, V_6.

ST-J depression of 1.0 mm or more and ST segment upward-sloping or U-shaped in any of leads I, aVL, V_6.

(2) Posterior (inferior) site (leads II, III, aVF):

ST-J depression 1.0 mm or more and ST segment horizontal or downward-sloping in any of leads II, aVF.

ST-J depression at least 0.5 mm and less than 1.0 mm and ST segment horizontal or downward-sloping in any of leads II, aVF.

No ST-J depression as much as 0.5 mm but ST segment downward-sloping and segment or T-wave nadir at least 0.5 mm below P-R baseline in lead II.

ST-J depression of 1.0 mm or more and ST segment upward-sloping or U-shaped in lead II.

(3) Anteroseptal site (leads V_1–V_5):

ST-J depression 1.0 mm or more and ST segment horizontal or downward-sloping in any of leads V_1–V_5.

ST-J depression at least 0.5 mm and less than 1.0 mm and ST segment horizontal or downward-sloping in any of leads V_1–V_5.

No ST-J depression as much as 0.5 mm but ST segment downward-sloping and segment or T-wave nadir at least 0.5 mm below P-R baseline in any of leads V_2–V_5.

ST-J depression of 1.0 mm or more and ST segment upward-sloping or U-shaped in any of leads V_1–V_5.

T-Wave Abnormality

(Do not code in presence of WPW syndrome, complete left bundle-branch block, complete right bundle-branch block, or intraventricular block with QRS duration of 0.12 sec or more in any of leads I, II, III, aVL, aVF.)

YES = If any of the T-wave abnormalities described in this section are present. Check location of T-wave abnormalities (anterolateral, inferior, or anteroseptal).

NO = If none of the T-wave abnormalities in this section are present.

(1) Anterolateral site (leads I, aVL, V_6):

T amplitude negative, minus 5 mm or more in any of leads I, V_6, or in lead aVL when R amplitude is 5 mm or more.

T amplitude negative or diphasic (positive-negative or negative-positive type), with negative phase at least minus 1.0 mm but not as deep as minus 5 mm in any of leads I, V_6, or lead aVL when R amplitude is 5 mm or more.

T amplitude zero (flat), negative, or diphasic (negative-positive type), with negative phase of less than 1.0 mm in any of leads I, V_6, or in lead aVL when R amplitude is 5 mm or more.

Optional code: T amplitude positive and T/R amplitude ratio less than 1/20 in any of leads I, aVL, V_6; R-wave amplitude must be 10 mm or more.

(2) Posterior (inferior) site (leads II, III, aVF):

T amplitude negative, minus 5 mm or more in lead II or lead aVF when QRS is mainly upright.

T amplitude negative or diphasic (positive-negative type), with negative phase at least minus 1.0 mm but not as deep as minus 5 mm in lead II or in lead aVF when QRS is mainly upright.

T amplitude zero (flat), negative, or diphasic (negative-positive type), with negative phase of less than 1.0 mm in lead II; not coded in lead aVF.

Optional code: T amplitude positive and T/R amplitude ratio less than 1/20 in lead II; R-wave amplitude must be 10 mm or more.

(3) Anteroseptal site (leads V_2–V_5):

T amplitude negative, minus 5 mm or more in any of leads V_2–V_5.

T amplitude negative or diphasic (positive-negative type), with negative phase at least minus 1.0 mm but not as deep as minus 5 mm in any of leads V_2–V_5.

T amplitude zero (flat), negative, or diphasic (negative-positive type), with negative phase of less than 1.0 mm in any of leads V_3–V_5.

Optional code: T amplitude positive and T/R amplitude ratio less than 1/20 in any of leads

V_3–V_5; R-wave amplitude must be 10 mm or more.

ST Segment Elevation

(Do not code in presence of WPW syndrome, complete left bundle-branch block, complete right bundle-branch block, or interventricular block with QRS duration 0.12 sec or more in any of leads I, II, III, aVL, aVF.)

YES = If ST segment elevation of 1.0 mm or more is present in leads I, II, III, aVL, aVF, V_1–V_6. Indicate ECG location of ST segment elevation.

NO = If no ST segment elevation of 1.0 mm or more is present.

(1) Anterolateral site (leads I, aVL, V_6): ST segment maximum elevation of 1.0 mm or more in any of leads I, aVL, or V_6.

(2) Posterior (inferior) site (leads II, III, aVF): ST segment maximum elevation of 1.0 mm or more in any of leads II, III, or aVF.

(3) Anteroseptal site (leads V_1–V_5): ST segment maximum elevation of 1.0 mm or more in lead V_5 or ST segment maximum elevation of 2.0 mm or more in any of leads V_1–V_4.

QRS Axis Deviation

(Do not code in the presence of WPW syndrome, complete left bundle-branch block, complete right bundle-branch block, or interventricular block, with QRS duration 0.12 sec or more in any of leads I, II, III, aVL, or aVF or low-amplitude QRS.)

YES = When one of the possibilities described in this section is present. (QRS axis $> +90°$ or more negative than $-29°$.)

NO = When none of the possibilities in this section are present. (QRS axis from $+89°$ to $-29°$.)

"Left" = QRS axis from $-30°$ through $-90°$ in leads I, II, III. (Algebraic sum of major positive and major negative QRS waves must be zero or positive in I, negative in III, and zero or negative in II.)

"Right" = (1) QRS axis from $+120°$ through $-150°$ in leads I, II, III. (Algebraic sum of major positive and major negative QRS waves must be negative in I, and zero or positive in III, and in I must be one-half or more of that in III.)

(2) QRS axis from $+90°$ through $+119°$ in leads I, II, III. (Algebraic sum of major positive and major negative QRS waves must be zero or negative in I and positive in II and III.)

"Extreme" (usually S1, S2, S3 pattern) = QRS axis from $-90°$ through $-149°$ in leads I, II, and III. (Algebraic sum of major positive and major negative QRS waves must be negative in each of leads I, II, and III.)

"Indeterminate" = QRS axis approximately 90° from the frontal plane. (Algebraic sum of major positive and major negative QRS waves is zero in each of leads I, II, and III, or the information from these three leads is incongruous.)

Ventricular Hypertrophy (High-Amplitude R Waves)

(Do not code in the presence of WPW syndrome, complete left bundle-branch block, complete right bundle-branch block, or interventricular block, with QRS duration 0.12 sec or more in any leads I, II, III, aVL, aVF.)

YES = If there is ECG evidence of ventricular hypertrophy (high-amplitude R waves) as defined below.

NO = If there is no ECG evidence of ventricular hypertrophy (high-amplitude R waves) as defined below.

"LVH" = (1) R amplitude greater than 26 mm in either of leads V_5 or V_6; or R amplitude greater than 20 mm in any of leads I, II, III, aVF; or R amplitude greater than 12 mm in lead aVL.

(2) R amplitude greater than 15 mm but less than 20 mm in lead I, or R amplitude in V_5 or V_6 plus S amplitude in V_1 greater than 35 mm.

"RVH" = R amplitude equal to or greater than 5.0 mm and R amplitude equal to or greater than S amplitude in lead V_1 when a decreasing R/S amplitude ratio occurs somewhere to the left of V_1 on the chest.

A-V Conduction Defect

YES = If atrioventricular conduction defect defined below is present.

NO = If no atrioventricular conduction defect is present.

"Complete" = Third-degree A-V block (permanent or intermittent) in any lead.

"Partial" = Second-degree A-V block in any lead (2:1 or 3:1 block, Wenckebach, etc.).

"P-R 0.22 sec" = P-R (P-Q) interval 0.22 sec or more in any of leads I, II, III, aVL, aVF.

"WPW" = WPW syndrome with P-R (P-Q) interval less than 0.12 sec, plus QRS duration 0.12 sec or more, plus R-peak duration 0.06 sec or more, coexisting in the same beats of any of leads I, II, aVL, V_4-V_6.

"Short P-R" = Short P-R (P-Q) interval with P-R (P-Q) interval less than 0.12 sec in all beats in any two of the following leads: I, II, III, aVL, aVF (in absence of A-V nodal rhythm and sinus tachycardia over 100/min).

"Artificial Pacemaker" = Self-evident.

Ventricular Conduction Defect

YES = If a ventricular conduction defect, as defined in this section, is evident from the ECG.

NO = If no ventricular conduction defect, as defined in this section, is evident on ECG.

"RBB" = Complete right bundle-branch block (in absence of WPW): QRS duration 0.12 sec or more in any of leads I, II, III, aVL, aVF *plus* R prime greater than R *or* R-peak duration 0.06 sec or more in either of leads V_1, V_2.

"LBB" = Complete left bundle-branch block (in absence of WPW): QRS duration 0.12 sec or more in any of leads I, II, III, aVL, aVF; and R-peak duration 0.06 sec or more and the absence of codable Q waves, in any of leads I, II, aVL, V_5, V_6.

"LAH" = Left anterior hemiblock: Left axis deviation (usually $>-60°$), small Q in lead I, small R in lead III, normal QRS duration.

"LPH" = Left posterior hemiblock: Right axis deviation (usually $>+120°$), small R in lead I, small Q in lead III, normal QRS duration, no evidence for RVH.

Arrhythmia

YES = If any arrhythmia present. (Pertains to the prevalent status at the time of this report.)

NO = If no arrhythmia present.

"Occasional PVC" = Less than 5 premature ventricular contractions per minute or less than 1 in 10 beats.

"Frequent PVC" = Five or more premature ventricular contractions per minute or more than 1 in 10 beats.

"Multifocal PVC" = Multifocal premature ventricular contractions. Mark also occasional PVC or frequent PVC as applicable.

"Ventricular Tachycardia" = Idioventricular rhythm at rate >100/min.

"Idioventricular" = Ventricular (idioventricular) rhythm of up to 100/min.

"Atrial Fibrillation or Flutter" = Readily understood.

"Supraventricular Tachycardia" = Supraventricular tachycardia at >100/min.

"Sinus Bradycardia" = <50/min.

"Nodal" = A-V nodal rhythm, rate up to 100/min. Defined as a negative P wave in lead aVF *plus* a P-R interval of 0.12 sec or less in any two of the following leads: I, II, III, aVL, or aVF.

"Other" = Arrhythmia not mentioned above. Please identify.

Miscellaneous

Low-Amplitude QRS

YES = Low QRS amplitude: QRS peak-to-peak amplitude less than 5 mm in each of leads I, II, III, or QRS peak-to-peak amplitude less than 10 mm in each of leads V_1-V_6.

NO = Low QRS amplitude not manifest.

QRS Transition Zone

Check precordial V lead where QRS transition occurs.

Electrocardiographic Stress Test

YES = Stress ECG test was done (either exercise and/or right atrial pacing).

NO = No ECG stress test was done.

Positive or Negative ECG

Record maximum heart rate per minute achieved during exercise and/or right atrial pacing stress test. Place this number in appropriate column to indicate whether stress test was "Negative" or "Positive."

"Positive ECG" = Stress ECG test with horizontal or downward-sloping ST segment depression of 1 mm or more or ST segment elevation of 1 mm was noted on three consecutive beats.

"Negative ECG" = Either normal or "negative" during or following stress ECG testing. This is an ECG tracing in which a 1-mm segment depression or elevation was not observed at the heart rate recorded.

If maximal heart rate achieved is not known, please simply check the appropriate box (whether positive or negative) for exercise stress testing or atrial pacing stress testing or for both if both were done.

Typical Angina Pectoris Induced

YES = Typical angina pectoris was induced during either/both the exercise stress ECG or/and the right atrial pacing tachycardia.

NO = No typical angina pectoris was induced during stress exercise ECG tests or right atrial pacing tachycardia.

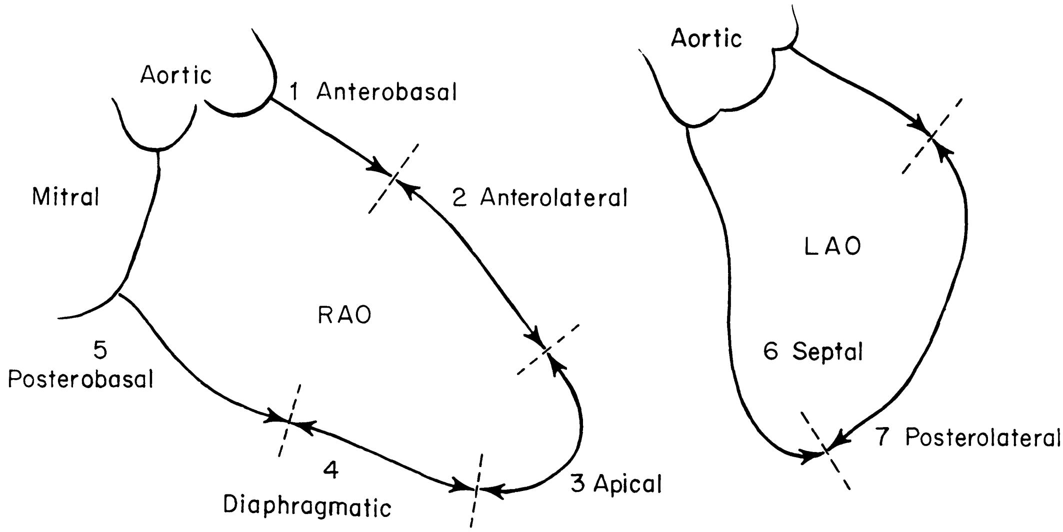
Other

Record other events that may have developed during stress ECG tests. Examples: marked dyspnea, ventricular arrhythmia, profound fatigue.

YES = If significant symptom other than typical angina pectoris was induced by stress ECG testing.

NO = If no other significant symptom was manifested during stress ECG testing.

LEFT VENTRICULOGRAM (LVG)



Note 1. Answer YES or NO to indicate whether patient received propranolol 48 hours before cardiac catheterization and whether patient was under effects of nitroglycerin at time of observation of left ventricular contractility and left ventricular end-diastolic pressure.
2. Check box to indicate whether LVG preceded or followed coronary arteriogram.

Contractility — As determined by left ventriculogram (LVG):

"*Normal*" = Normal wall motion of indicated ventricular segment.

"*Reduced*" = Velocity and/or amplitude of indicated wall ventricular segment motion are/is reduced.

"*None*" = Absence of appropriate wall motion of indicated ventricular segment.

"*Dyskinetic*" = Paradoxical wall motion of indicated ventricular segment.

"*Aneurysmal*" = Aneurysmal bulging with sharply defined margins of indicated ventricular segment.

"*Undefined*" = Status of indicated ventricular wall segment cannot be described because of improper visualization or omission of appropriate views.

HEMODYNAMICS

Values to be recorded as follows:

Aortic Pressure:	Systolic and diastolic pressure in mm Hg.
Heart Rate:	Average heart rate at time of study.
LV Pressure:	Left ventricular systolic and end-diastolic pressures.
Cardiac Index:	Cardiac index (liters/min/m^2).
LV End-Diastolic Volume:	Left ventricular end-diastolic volume index (ml/m^2).
Ejection Fraction:	Ejection fraction (%).
VSD:	Record whether a ventricular septal defect is present (YES or NO).

Pul/Systemic Flow Ratio: Ratio of pulmonary to systemic blood flow (examples: 1.2/1, 2/1, etc.).

Mitral Valve Gradient: Mean left atrial pressure (or mean pulmonary artery wedge pressure) minus left ventricular end-diastolic pressure.

Aortic Valve Gradient: Peak systolic aortic valve gradient in mm Hg.

Mitral Regurgitation: Mitral valve regurgitation.
"None" = No evidence of mitral regurgitation whatsoever (except obviously artifactual regurgitation).
"Slight" = Mitral regurgitation present but barely detectable.
"Moderate" = Obvious evidence of mitral regurgitation resulting in detectably uniform opacification of left atrium.
"Severe" = Mitral regurgitation resulting in prompt and intense opacification of left atrium even before the ascending aorta is fully visualized.

Aortic Regurgitation: Aortic valve regurgitation.
"None" = No evidence of aortic valve regurgitation whatsoever except that which is obviously artifactual.
"Slight" = Aortic regurgitation is present but barely detectable. Left ventricle is never completely opacified.
"Moderate" = Obvious aortic regurgitation resulting in uniform opacification of the left ventricle. There is prompt clearing of contrast from the left ventricle after injection is complete.
"Severe" = Aortic valve regurgitation resulting in prompt and intense opacification of the left ventricle. This opacification persists for several beats after injection of contrast has stopped.
"No aortogram" = An aortogram was *not* done during catheterization.

CORONARY ARTERIOGRAM

Note: Record the most severe narrowing of the indicated coronary arterial segment if more than one narrowing is present in a particular segment.

Record as follows:

"Normal": Indicated segment is entirely within normal limits.

"Gives Collateral": Indicated segment is *giving* collaterals to some other vessel or branch.

"Small": Indicated vessel or segment of vessel is small throughout either because of anatomic variation or diffuse narrowing. Score both "Normal" and "Small" if normal *and* small; score both "Small" and degree of narrowing if disease is diffuse.

"25%": Lumen diameter reduction up to but not exceeding 25%.

"50%": Lumen diameter reduction from 26% up to but not exceeding 50%.

"75%": Lumen diameter reduction from 51% up to but not exceeding 75%.

"90%": Lumen diameter reduction from 76% up to but not exceeding 90%.

"Filled by Collaterals": Indicated vessel or segment of vessel is being opacified by collateral channels.

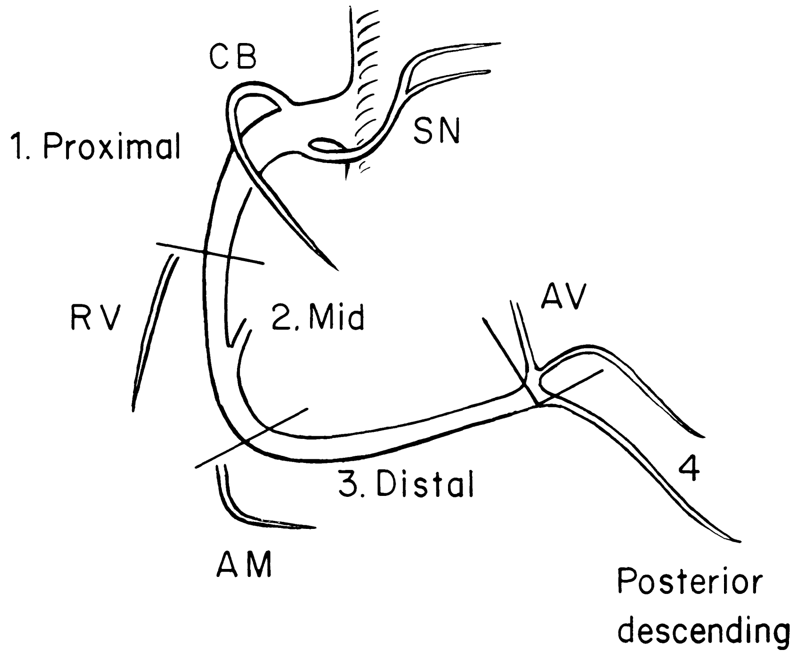
"99%": Lumen appears nearly obliterated or reduced to hair-width (greater than 90% narrowing) but some contrast still passes by this narrowing.

"100%": Complete occlusion of indicated branch or segment of vessel.

"Graftable": Segment of vessel that either appears angiographically suitable for graft anastomosis (when arteriogram is a preoperative study) or where graft was placed (if arteriogram is a postoperative study).

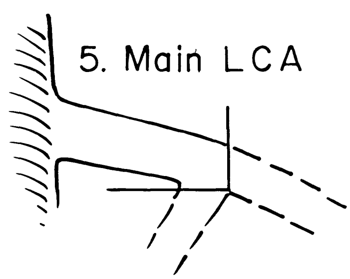
"Absent": Indicated segment is absent due to *unquestionable* anatomic variation.
Note: 1. Not to be checked when vessel may be absent because of proximal occlusion.
2. This should also be checked for the first segment of a major artery when there is ectopic origin of that artery (e.g., origin of circumflex from right coronary artery). Site of anomalous origin of artery should also be written in.

RIGHT CORONARY ARTERY (RCA)



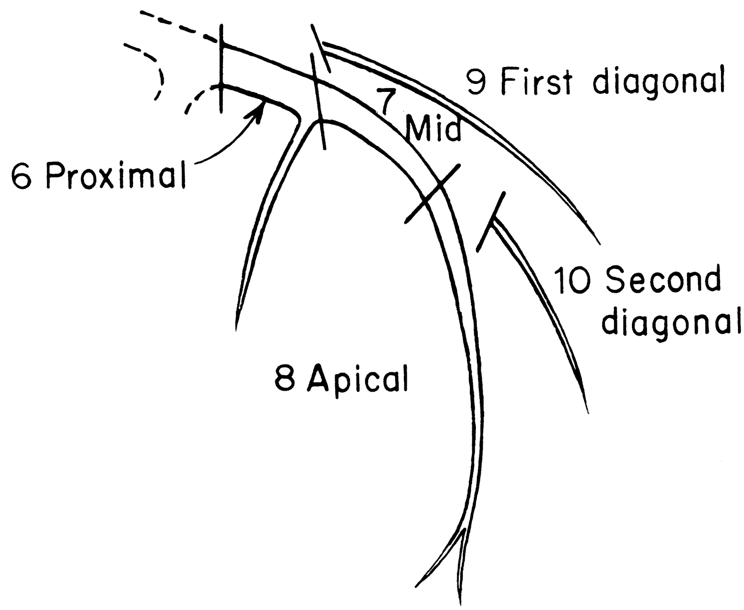
1. Proximal	RCA from ostium to one-half the distance to the acute margin of heart. *Note:* Distal portion of this coronary segment may coincide with origin of one of the large right ventricular branches (RV).
2. Mid	RCA from end of above segment to acute margin of heart. *Note:* Distal portion of this segment *usually* (but not always) coincides with origin of the acute marginal branch (AM).
3. Distal	RCA running along the posterior right atrioventricular groove, from the acute margin of heart to the origin of the posterior descending branch. *Note:* In the "predominant" left coronary pattern (posterior descending from left circumflex), this segment may be very small.
4. Posterior Descending	Posterior descending branch, when present. *Note:* Minor branches such as the conus (CB); sinus node (SN); right ventricular (RV); acute marginal (AM); and A-V node (AV) branches are indicated in diagram only for general orientation. These branches may or may not be visualized in the individual patient. Those that are highly variable as to their origins are shown unattached to parent artery.

MAIN LEFT CORONARY ARTERY (LCA)



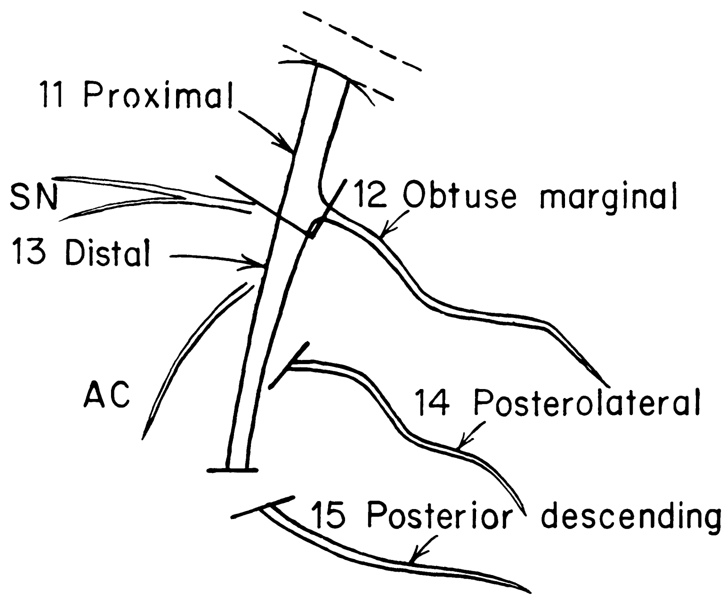
5. Main Left	LCA extends from the ostium of LCA through bifurcation into left anterior descending and left circumflex branches.

LEFT ANTERIOR DESCENDING ARTERY (LAD)



6. Proximal	LAD proximal to and including origin of the first major septal perforator branch.
7. Mid	LAD immediately distal to origin of first major septal perforator branch and extending to point where LAD forms an angle (RAO view) and often but not always coinciding or close to the origin of second diagonal branch (D_2). If this angle or branch is not identifiable, this segment ends at one-half the distance from the first major septal perforator branch to apex of heart.
8. Apical	Terminal portion of LAD running along interventricular sulcus, beginning with the end of previous segment, and usually extending beyond apex.
9. First Diagonal	First diagonal (D_1): The largest, and usually the first, diagonal branch having its origin from proximal segment of LAD. Occasionally, this can be a separate branch off main LCA.
10. Second Diagonal	Second diagonal (D_2): The second diagonal branch, which often has its origin at the "angle" of LAD, descending when visualized in RAO projection.

LEFT CIRCUMFLEX CORONARY ARTERY (CIRC)



11. Proximal	Main stem of circumflex from its origin off main LCA to and including origin of obtuse marginal branch.
12. Obtuse Marginal (OM)	The largest branch of circumflex running in general area of obtuse margin of heart.
13. Distal	The stem of the circumflex distal to origin of obtuse marginal branch and running along or close to posterior left atrioventricular groove. Caliber of this portion may be small.
14. Posterolateral (PL)	Posterolateral branch of circumflex distributing to posterolateral surface of left ventricle; almost always smaller in caliber than obtuse marginal. May be absent or may be a division of obtuse marginal branch.
15. Posterior Descending (PD)	Posterior descending, when present as a branch of circumflex.

Note: Minor or variable branches such as the sinus node (SN) and the atrial circumflex (AC) are indicated in the diagram only for the purpose of general orientation. They may or may not be visualized in the individual patient.

Angiography of Coronary Bypass Graft

This section records the angiographic appearance of the coronary artery bypass graft(s) at some time after operation. Each graft is identified by the segment of coronary artery into which it is inserted. It is assumed that the type of graft, and the origin of each graft, has been previously described on the *Surgical Report*.

Record using as guides the following: (1) Is bypass graft patent in its entirety? (2) Is bypass graft totally occluded in any portion? (3) If the graft appears "Normal," check appropriate column. (4) "Irregularly Narrowed (<50%)" describes the bypass graft with recognizable narrowings that are less than 50% and are not considered to appreciably impair blood flow through this graft. (5) "Diffusely Narrowed (>50%)" describes the bypass graft with appreciable narrowing throughout most of its length that is considered to impair blood flow through this graft. (6) If there is a recognized "Aneurysmal" dilatation of the bypass graft, check appropriate column. (7) To define more accurately the degree of luminal occlusion of the most severe narrowing of each bypass graft, estimate whether the diameter of the lumen at this narrowing is: (a) <1 mm (90% occluded); (b) 1 to <2 mm (75% occluded); (c) 2 to <3 mm (50% occluded); (d) 3 to <4 mm (25% occluded); or (e) >4 mm (normal). (8) Finally, record a check if the bypass graft was visualized by "Selective Injection" of the graft. (9) Check "Unknown" if no information is available concerning this bypass graft.

Recommended Medical Therapy

The type of medical therapy that was recommended to a patient on the basis of these data should be recorded. Dosage and mode of administration are not included. If no medical therapy was recommended, so indicate. If operation was recommended but refused, record medical therapy then recommended. If the reason for this entry is an operation for coronary arterial disease, and subsequent medical therapy is recommended, so indicate.

Recommended Surgical Therapy

Patients may receive conflicting advice. The recommendation to be included here is that given by the responsible physician and surgeon at the institution from which this report is made; it has been based on the data here recorded. Please complete the appropriate box.

If the reason for the entry is cardiac operation, the status of the patient immediately before operation forms the basis for the *Clinical and Arteriographic Report;* the *Surgical Report* must also be completed. (See below for explanation and instructions for *Surgical Report.)*

Patient Death

If the reason for the entry is death of the patient, the status of the patient just before death forms the basis for the report. If autopsy was performed, cardiac findings at autopsy should be entered into the *Autopsy Report.* (See also explanations and instructions for *Autopsy Report.)*

SURGICAL REPORT

Explanation and Instructions

Patient Identification

Enter patient's name, social security number, date of operation, and name of primary surgeon. (The primary surgeon is defined as the surgeon who, in the patient's understanding, performed his operation.) Indicate whether primary surgeon is a Staff Surgeon or Resident.

If the patient's status remains the same as that at the time of the last previous entry (usually same as entry including coronary arteriogram), and if the interval since that entry is less than 6 months, only the *Surgical Report* need be completed; a check should be made to indicate "Previous Entry Applies." However, if the patient's status has changed from the time of the last entry or if longer than 6 months have elapsed, a new *Clinical and Arteriographic Report* should be completed to accompany this *Surgical Report* and "Accompanied by New Entry" should be checked. Record date of most recent preoperative coronary arteriogram.

Operative Procedure

The bypass grafts are identified by the coronary artery segment into which they are inserted. The coronary segments are exactly as defined under the instructions for the *Clinical and Arteriographic Report* (pp 31-33). All data that are available for each graft should be filled in on the line adjacent to the coronary artery segment into which this graft was inserted.

"*Type of Bypass Graft.*" — Record a number in the appropriate box according to the following key:

1 = Saphenous vein autograft
2 = Other venous autograft
3 = Right internal mammary artery (IMA)
4 = Left internal mammary artery
5 = Other arterial autograft
6 = Venous homograft
7 = Arterial homograft
8 = Prosthetic artery
9 = Arterial heterograft
10 = Other

"*Origin of Bypass Graft.*" — The site of origin of each bypass graft should be recorded in the appropriate box according to the following key:

A = Aorta
B = Natural origin for that artery
C = Side branch of another graft
D = Continuation of graft distal to side-to-side anastomosis to other coronary segment
E = Other

For example, the patient who has a left internal mammary artery bypass graft into the distal left anterior descending artery (segment 8) and a saphenous vein to the right coronary artery (segment 3) would be recorded as follows: adjacent to "LAD segment 8" record 4-B, adjacent to "RCA segment 3" record 1-A.

"*Endarterectomy.*" — If an endarterectomy of the coronary segment into which the graft was inserted was performed, indicate by marking the appropriate box.

"*Quality Coronary Artery.*" — Describe and record coronary artery segment into which the graft is inserted as either normal, diseased, or occluded.

"Normal" = No atheromatous plaque or only mild intimal changes noted
"Diseased" = If definite atheromatous plaque is present
"Occluded" = If no lumen or a small recanalized lumen is present

"*Diameter Coronary Lumen.*" — The internal diameter of the coronary segment into which the graft is inserted should be recorded in millimeters. This is usually estimated by passing calibrated probes.

"*Aortic Cross-Clamp Time.*" — If the aorta was cross-clamped while the graft to artery anastomosis was made, record in minutes the duration of aortic cross-clamping required for each graft. This assumes that the clamp was released between insertion of multiple grafts. If the clamp was not released between grafts, place check in this column for each graft, and record the total time of cross-clamping on the line adjacent to the segment into which the last graft was inserted.

"*Graft Flow.*" — Record the blood flow (in ml/min) through each graft measured after completion of cardiopulmonary bypass while in a steady state and without deliberate ischemic or pharmacologic dilatation of the peripheral coronary circulation.

"*Unusual Problem.*" — If something unusual developed or was required during the procedure, such as revising an anastomosis or repairing the back wall of the coronary artery, indicate succinctly.

Ventricular Fibrillation. — If ventricular fibrillation did not develop during the operative procedure, mark accordingly in box for "None." If it did, indicate if this was induced electrically or developed spontaneously. If ventricular fibrillation resulted from causes other than purposeful electrical inducement, such as from myocardial cooling, mark as though it developed spontaneously. If, after completion of surgery, one or more electrical countershocks were required for conversion to a coordinated rhythm, so indicate by checking "Countershock Required."

Temperature of Myocardial Perfusate. — Indicate appropriately the range of temperature of the perfusate supplying the myocardium during the cardiac procedure or immediately preceding the longest period that the aorta was cross-clamped. Unless a special system was incorporated in the heart-lung machine, this temperature would be the same as for the perfusion of the whole body.

Surface Cardiac Cooling. — If a technique was used for cooling the surface of the heart by any means other than ambient room air, indicate by checking YES, otherwise check NO.

Duration of Extracorporeal Circulation. — The total time in minutes of partial and complete cardiopulmonary bypass should be recorded.

Other Cardiac Procedures

If some other cardiac procedure was combined with coronary artery bypass or was performed without coronary artery bypass, so indicate.

Valve. — If the aortic or mitral valve, or both, was replaced or repaired, indicate appropriately.

Ventricular Segment. — If a portion of the left ventricular wall was resected or plicated, so indicate. The ventricular segment that was resected or plicated should be indicated by recording the number(s) corresponding to the ventricular segment(s) depicted in the section on Left Ventriculogram of *Clinical and Arteriographic Report* (p 29).

Resected Segment. — The pathologic description of the resected segment of left ventricular myocardium is appropriately noted by checking one of the five options. The box to be checked is determined by reference to the following definitions: "Aneurysm," a definite outward bulging of the cardiac contour in the area of the infarction; "Acute Infarct," an infarct estimated to be less than 1 week old; "Healing Infarct," an infarct 1 week to 1 month old; "Healed Infarct," an infarct older than 1 month; and "Fibrosis," intermingling of residual myocardial tissue and focal fibrous changes.

Ventricular Septal Defect Repair. — If a ventricular septal defect was repaired, so indicate.

Other Operative Procedures. — If some other cardiac operative procedure was combined with coronary artery bypass or was performed without accompanying coronary artery bypass, so indicate.

Postoperative Course

If the interval from operation to discharge from the hospital was free of significant complications, check "No Complication," otherwise check "Complication." Record any ECG change(s) and the highest values of serum enzymes after operation, even though the postoperative course was considered to be consistent with "No Complication."

Electrocardiographic Changes. — If the ECG after operation showed any appreciable change from the preoperative tracing, check YES, otherwise check NO. These changes include the appearance of "New Q or QS Pattern," "New ST Segment Shift" (either elevation or depression, both of which may be transient), and "New T-Wave Inversion" (which may be transient). The definitions for these three changes and the location of these changes are given in the instruction booklet for the *Clinical and Arteriographic Report* (pp 25-26).

Any change in cardiac rhythm (such as the appearance of atrial fibrillation) should be recorded under Morbidity (see below).

Serum Enzyme Determinations. — Record the maximum value found during the postoperative hospital period for any or all of four serum enzyme determinations. These serum enzymes are: serum glutamic-oxalacetic transaminase (SGOT), lactic dehydrogenase (LDH), total creatine phosphokinase (CPK), and creatine phosphokinase-MB isozyme (CPK-isozyme).

Also record the upper limit of normal at that institution (or above the 95th percentile of test results if institution employs this system) for each of these serum enzymes. If a serum enzyme was not determined after operation, check appropriate box.

Morbidity. — Indicate by checking YES or NO whether any of the following problems occurred during the postoperative in-hospital period:

"Mediastinal Infection." If an infection occurred postoperatively in the retrosternal area, check YES. If no infection occurred in this area, check NO. Other infections such as those of urinary tract, respiratory tract, presternal incision, or donor vein site are not to be considered for this entry item.

"Pulmonary Embolism." If pulmonary embolism was diagnosed with a high degree of probability on clinical grounds or was documented by angiography, check YES. If no pulmonary embolism was considered to have occurred, check NO.

"Low Cardiac Output." For the purposes of this report, low cardiac output is considered to be the result of impaired left ventricular performance and may be manifest as either hypotension, poor organ perfusion, or fluid retention. Check YES if a state of low cardiac output occurred sufficiently to require specific therapy with an inotropic agent or agents, despite correction of blood volume, cardiac rhythm, acid-base balance, or other preceding cause. Also, check YES if there is clinical evidence of appreciable fluid retention (e.g., elevated central venous pressure, pulmonary congestion or edema, hepatomegaly, weight gain) that has to be treated by specific diuretic therapy.

If there is no evidence of low cardiac output or fluid retention, check NO.

"Excessive Hemorrhage." Excessive postoperative bleeding is arbitrarily defined as more than 1,000 ml bloody drainage from the chest tubes in the first 24 postoperative hours. Check YES or NO appropriately.

"Reoperation for Hemorrhage." If reoperation for bleeding was required, check YES, otherwise check NO.

"Arrhythmia." If a new and significant arrhythmia or conduction defect occurred after operation — between the time that cardiopulmonary bypass was completed and patient was discharged from hospital — check YES. If no cardiac arrhythmia of significance occurred during the postoperative period or if this arrhythmia was present before operation (e.g., atrial fibrillation, frequent premature ventricular contractions) as well as after surgery, then check NO.

Indicate the type of cardiac arrhythmia or conduction defect that occurred by checking the appropriate box or writing in the specific arrhythmia after the word "Other." (The definitions of these cardiac arrhythmias and conduction abnormalities are given in the instruction booklet for *Clinical and Arteriographic Report* [pp 28].) The following guide should be used:

a. Atrial fibrillation or flutter. If atrial fibrillation or flutter appeared after operation (not present immediately before operation), mark box.

b. Supraventricular tachycardia. If one or more episodes of supraventricular tachycardia occurred, so indicate (includes paroxysmal atrial tachycardia).

c. Frequent premature ventricular contractions (PVC). If PVC's were frequent (i.e., five or more per minute or more than two in 10 beats) during the postoperative course, mark box. Specific suppressive therapy with antiarrhythmic drugs such as xylocaine, quinidine, and procainamide may or may not have been administered.

d. Ventricular tachycardia or fibrillation (VT/VF). If ventricular tachycardia (an idioventricular rhythm at a rate exceeding 100/min) or ventricular fibrillation occurred in the postoperative period (beginning after completion of cardiopulmonary bypass), mark box.

e. Other. If any other significant arrhythmia or conduction abnormality occurred, check box and write in specific diagnostic term.

Hospital Mortality. — If the patient died in the hospital at any time after induction of anesthesia for cardiac surgery had begun, check YES, otherwise check NO. If the patient died, record postoperative interval between operation and death (in days). If the patient died during induction of anesthesia, did not come off cardiopulmonary bypass, or died within the first 24 hours after operation, record a zero (0) in the blank after "Days After Operation." Otherwise, record the day(s) after operation of death (e.g., day 3, etc.).

Check the appropriate option for the most important underlying cause of death. If the cause of death was other than those listed, record cause after "Other"; if unknown write unknown after "Other." In this instance, "Infection" includes infection at any site, whether in the retrosternal wound or at some other area of the body or bloodstream.

Autopsy Performed. — If autopsy was performed, check YES and complete *Autopsy Report;* if not performed, check NO.

AUTOPSY REPORT

Explanation and Instructions

Left Ventricle

Pathologic Changes in Segments. — The segments are defined exactly as in the section on Left Ventriculogram of the *Clinical and Arteriographic Report* (p 29). Segments 2, 3, and 6 reflect predominantly the vascular distribution of the left anterior descending coronary artery; segment 4, that of the right coronary artery; and segments 5 and 7, that of the circumflex coronary artery. Each segment should be identified as being normal or showing one of the designated pathologic changes and checked in the appropriate column and space. Obviously, any changes will not always follow the distribution of these defined segments, so that if a process in one segment overlaps for about 1/3 or more into another segment, the latter segment is marked as being involved also.

If an infarct is estimated to be less than 1 week old, a check should be made in the column for "Acute"; if 1 week to 1 month old, at "Healing"; and if older than 1 month, at "Healed." Also indicate whether the infarct is "Subendocardial" only or "Transmural." "Fibrosis" is checked to indicate that both residual myocardial tissue and focal fibrous changes are intermingled. "Aneurysm" is meant here to indicate a distinct outward bulging of the cardiac contour in the area of infarction.

Ventricular Septal Defect. — If any VSD is present, circle YES.

Aortic and Mitral Valves. — If stenosis is present, or more than mild regurgitation estimated to be present, so indicate.

Heart and Lung Weights. — If possible, record as measured weight over estimated normal for patient of that age and weight.

Fibrous Pericarditis. — If the pericardial space is obliterated by excessively fibrous adhesions that are significantly more dense than are consistent with the usual postoperative or postinfarct changes (if appropriate), circle YES.

Mediastinum. — Mark appropriately if infection is present, if hematoma is present in excess of any normal postoperative change (if appropriate), or if normal.

Coronary Arteries

Postmortem coronary angiography is highly desirable; indicate whether this was done. If not, it is assumed that the coronary arteries have been examined in minute detail at intervals of 3 mm or less. Definitions for this section are identical to those for the coronary arteriogram of the *Clinical and Arteriographic Report* (pp 30-33), but the two main columns relating to collateral vessels and the one indicating graftability should not be completed unless postmortem coronary arteriography was performed.

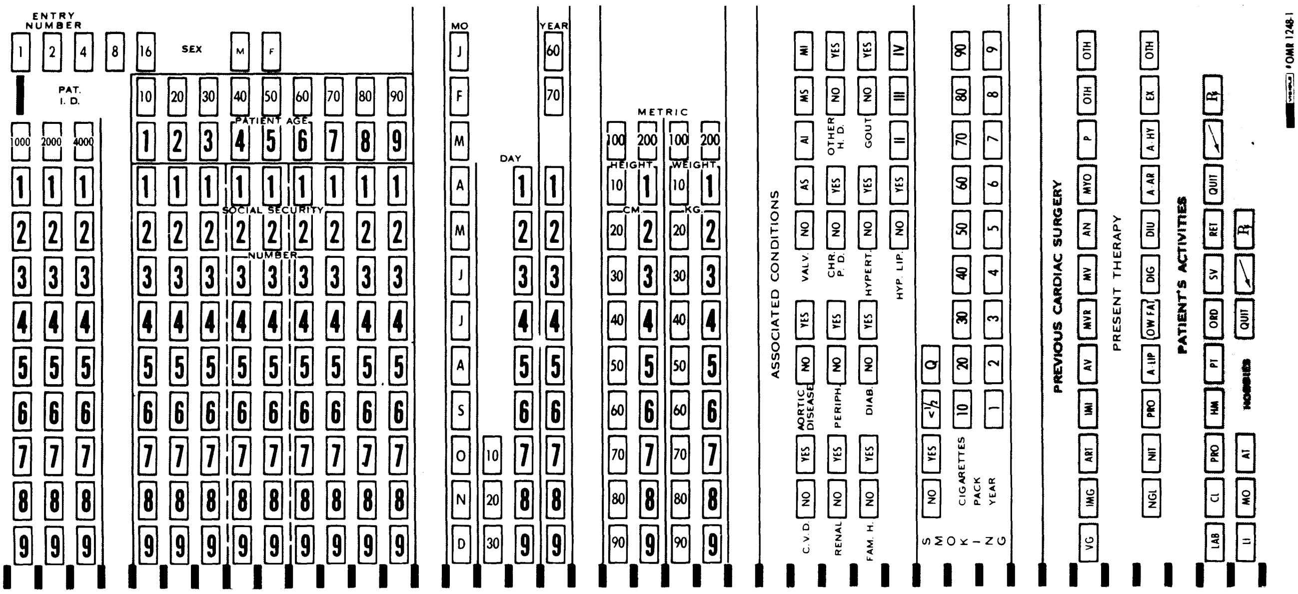
Coronary Bypass Graft. — If the patient had undergone coronary bypass grafting of one or more coronary arteries, this section should be completed exactly as explained for the *Clinical and Arteriographic Report* (p 33).

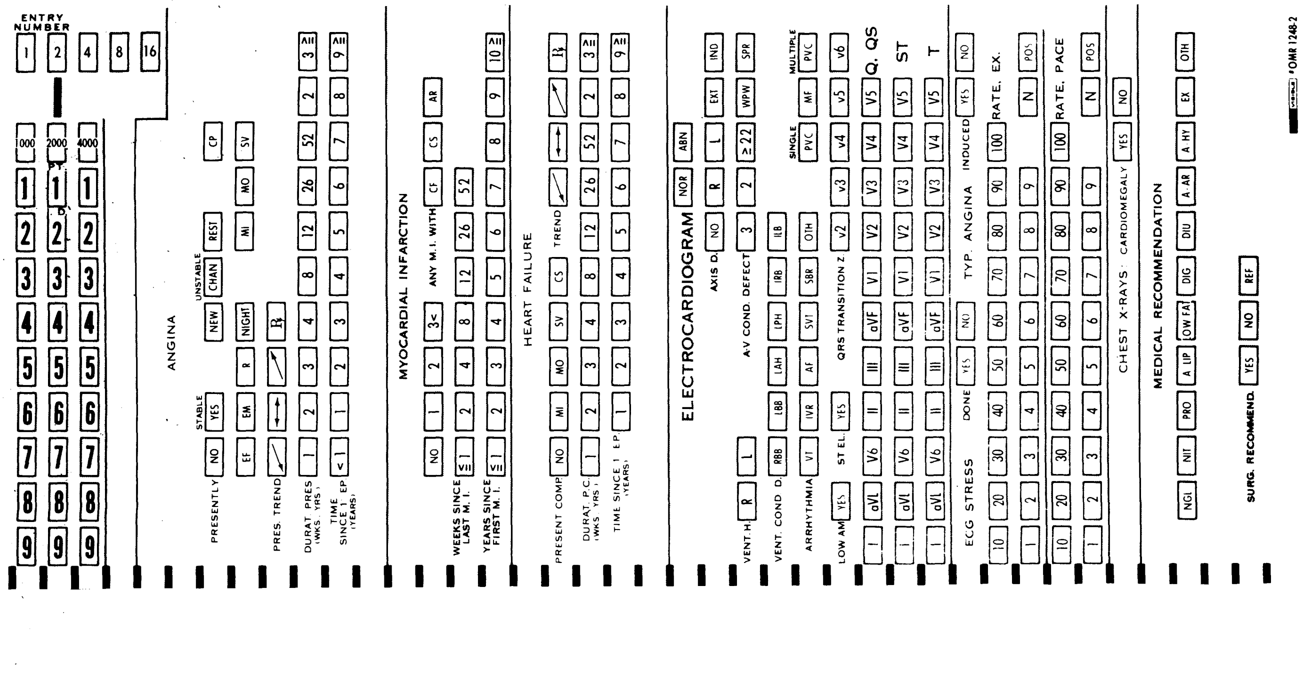
Cause of Death

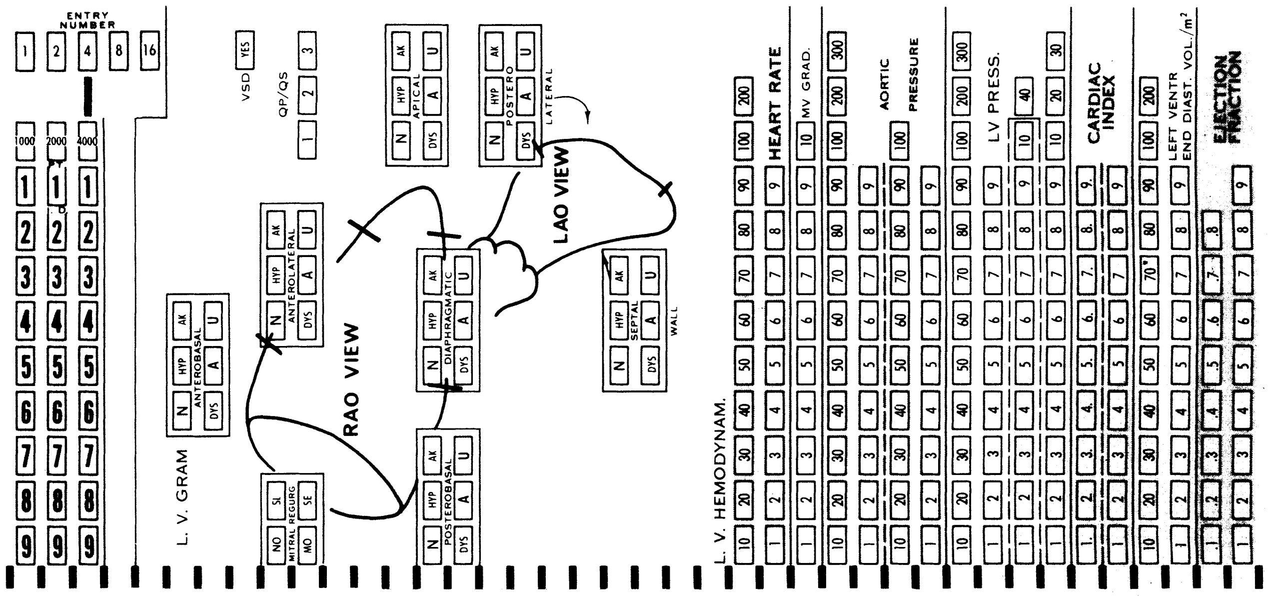
If death was considered to be directly related to a cardiac cause, check appropriately whether due to coronary or noncoronary disease; or if underlying cause of death was not primarily the result of cardiac disease, so indicate. Finally, write in the underlying cause of death.

Appendix

For full implementation of this Reporting System, computer assistance is indicated. The set of computer-compatible entry cards reproduced here represents an example of a scheme whereby computer interface can be accomplished.



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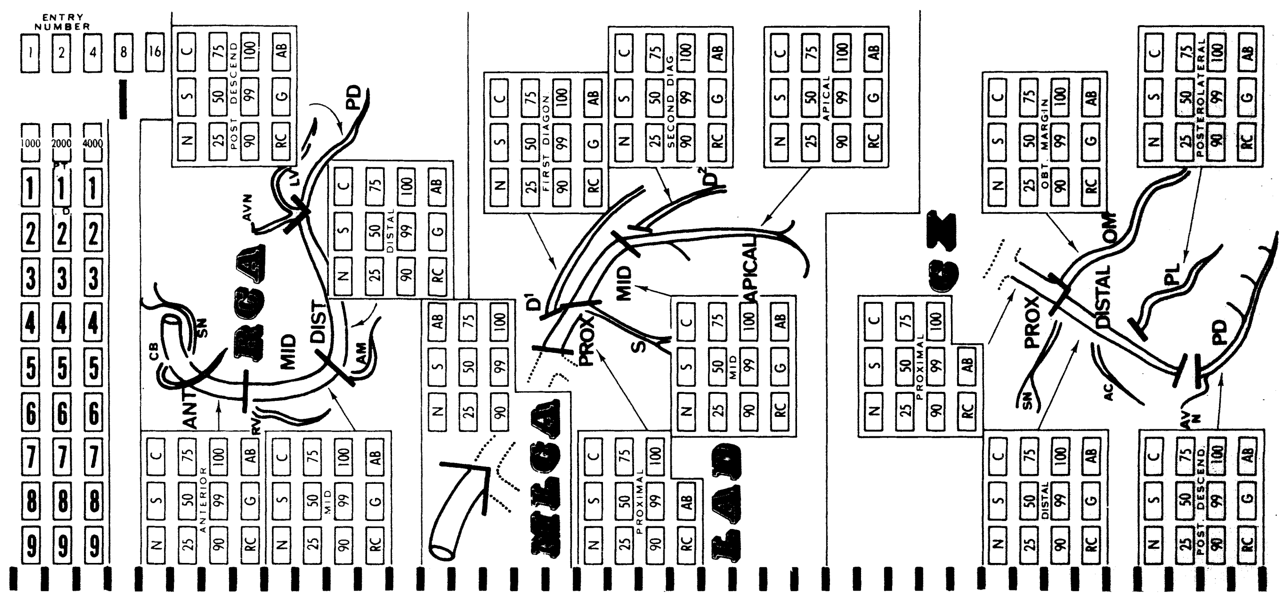
VSD YES
QP/QS
ANTEROBASAL
ANTEROLATERAL
APICAL
POSTEROBASAL
DIAPHRAGMATIC
POSTERO LATERAL
SEPTAL
WALL
MITRAL REGURG
RAO VIEW
LAO VIEW
L. V. HEMODYNAM.
HEART RATE
MV GRAD.
AORTIC
PRESSURE
LV PRESS.
CARDIAC INDEX
LEFT VENTR END DIAST. VOL./m²
EJECTION FRACTION
OMR 1248-3


ENTRY NUMBER
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MID
POST DESCEND
DISTAL
MLCA
LAD
PROXIMAL
FIRST DIAGON
SECOND DIAG
APICAL
CX
OBT. MARGIN
POSTEROLATERAL
POST. DESCEND.
OMR 1248-4